

Coding LLMs from the Ground Up: A Complete Course



SEBASTIAN RASCHKA, PHD

MAY 10, 2025 · PAID



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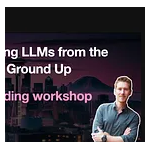
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I wrote a lot about reasoning models in recent months (4 articles in a row)! Next to everything "agentic," reasoning is one of the biggest LLM topics of 2025.

This month, however, I wanted to share more fundamental or "foundational" content with you on how to code LLMs, which is one of the best ways to understand how LLMs work.

Why? Many people really liked and benefited from the abbreviated LLM workshop shared last year:



Building LLMs from the Ground Up: A 3-hour Coding Workshop

SEBASTIAN RASCHKA, PHD · AUGUST 31, 2024

[Read full story →](#)

So, I thought this ~5× longer and more detailed content (~15 hours in total) would even more useful.

Also, I'm sadly dealing with a bad neck injury and haven't really been able to work on my computer for the past 3 weeks. I am currently trying a conservative treatment before considering the suggested surgical route. This is the worst timing as I just started to get back on track before life threw another curveball.

So, during my recovery, I thought sharing these videos I recorded in the last couple months would be a nice in-between content.

I hope you find this useful, and thanks for your support!

PS: The videos originally started as supplementary content for my [Build a Large Language Model \(From Scratch\) book](#). But it turns out they also work pretty well as standalone content.

Why build from scratch?

It's probably the best and most efficient way to learn how LLMs really work. Plus, many readers have told me they had a lot of fun doing it.

To offer an analogy: if you are into cars and want to understand how they work, following a tutorial that walks you through building one from the ground up is a great way to learn. Of course, we probably wouldn't want to start by building a Formula race car since it would be prohibitively expensive and overly complex for a first project. Instead, it makes more sense to start with something simpler, like a go-kart.

Building a go-kart still teaches you how the steering works, how the motor functions and more. You can even take it to the track and practice (and have a lot of fun with) before stepping into a professional race car (or joining a company or team that is focused on building one). After all, the best race drivers often got their start by building and tinkering with their own go-karts (think Michael Schumacher and Ayrton Senna). By doing that, they not only developed a great feel for the car but could also provide valuable feedback to their mechanics, which gave them an edge over the other drivers.

References

1. Build an LLM from Scratch book ([Manning](#) | [Amazon](#))
2. [Build an LLM from Scratch GitHub repository](#)

1 - Set up your code environment (0:21:01)

This is a supplementary video explaining how to set up a Python environment using uv.

In particular, we are using “uv pip”, which is explained in [this document](#).

Alternatively, the native “uv add” syntax (mentioned but not explicitly covered in this video) is described [here](#).



Build an LLM from Scratch 1: Set up your code environment

Sebastian Raschka



前往平台观看：

Note / Tip: The installation may cause issues on certain versions of Windows. If you are on a Windows machine and have troubles with the installation (likely due to a TensorFlow dependency to load the original GPT-2 model weights from OpenAI in video 5), please don't worry about it and feel free to skip the TensorFlow installation (you can do this by removing the TensorFlow line from the requirements file.)

To provide an alternative, I converted the GPT-2 model weights from a TensorFlow tensor format to PyTorch tensors and shared them on the Hugging Face model hub which you can use as an alternative to the weight loading portion in video 5: <https://huggingface.co/rasbt/gpt2-from-scratch-pytorch>.

In any case, you don't have to worry about this weight-loading code until the end of video 5.

2 - Working with text data (1:28:01)

This video goes over text data preparations steps (tokenization, byte pair encoding, data loaders, etc.) for LLM training.



Build an LLM from Scratch 2: Working with text data
Sebastian Raschka



前往平台观看：

3 - Coding attention mechanisms (2:15:40)

This is a supplementary video explaining how attention mechanisms (self-attention, causal attention, multi-head attention) work by coding them from scratch. You can think of it as building the engine of a car (before adding the frame, seats, & wheels).



Build an LLM from Scratch 3: Coding attention mechanisms

Sebastian Raschka



前往平台观看：

4 - Set up your code environment (0:21:01)

This video covers how to code an LLM architecture from scratch.



Build an LLM from Scratch 4: Implementing a GPT model from Sc

Sebastian Raschka



前往平台观看：

5 - Pretraining on Unlabeled Data (2:36:44)

This video explains how to pretrain a LLM from scratch.



Build an LLM from Scratch 5: Pretraining on Unlabeled Data

Sebastian Raschka



前往平台观看：

6 - Finetuning for Classification (2:15:29)

This is a video explaining how to fine-tune an LLM as a classifier (here using a spam classification example) as a gentle introduction to finetuning, before instruction finetuning the LLM in the next video.



Build an LLM from Scratch 6: Finetuning for Classification

Sebastian Raschka



前往平台观看：

7 - Instruction Finetuning (1:46:04)

Finally, this video explains how to instruction finetune the LLM.



Build an LLM from Scratch 7: Instruction Finetuning

Sebastian Raschka



前往平台观看：

Happy viewing & tinkering!

Bonus: LLMs Then And Now (From 2018 to 2025)

As a big thank you to the paid subscribers, I want to share a 2.5h (non-coding) bonus video I recorded earlier in April, approximately 2 days after the Llama 4 release. In the talk, I discuss the current LLM landscape in 2025 with a focus on what and how things have changed since GPT-2 in 2018.

Thanks for your support, as an independent and self-employed researcher, this really means a lot to me!

Hopefully, things will improve in the next few weeks/months as I have lots of ideas for upcoming articles and can't wait to work on them!

0:00



In addition, you can find a link to the slides here:

- <https://sebastianraschka.com/pdf/slides/2025-04-llms-whats-next.pdf>

And below is a table of contents for the video:

- 00:00 LLM Specialization
- 17:24 LLM Architecture changes
- 22:42 Mixture of Experts (MoE)
- 32:35 Multi-Head Latent Attention (MLA)
- 36:47 Model sizes over the years
- 43:39 Pre-training
- 50:09 Hardware
- 58:04 Training complexity
- 1:09:01 Pre-training datasets

- 1:24:22 Instruction fine-tuning
- 1:34:08 Preference tuning
- 1:46:17 Reasoning specialization
- 1:58:18 Different ways to build reasoning models
- 2:06:14 DeepSeek case study
- 2:19:21 Affordable reasoning model experiments

I hope you found this video useful, and many thanks for your support!



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Saumya 🌟 Saumya Jun 25, 2025

❤️ Liked by Sebastian Raschka, PhD

Hey Sebastian! hope you are recovering well.

Thanks for all your insightful detailed posts, and looking forward to learning more whenever yc back to 100%

I purchased the subscription today and it does not look like the paid subscriptions are paused

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1 reply by Sebastian Raschka, PhD



Hunan Rostomyan  Hunan Rostomyan May 10, 2025

 Liked by Sebastian Raschka, PhD

Hope the non-surgical approaches work Sebastian. Either way, speedy recovery. Thanks for all y
amazing work, learned a lot from you and looking forward to learning more whenever you're ba
100%

 LIKE (1)  REPLY



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